COMICCRAFT

Project Design Phase

AI Comic Story Creator – Project Phase Documentation

This document contains 10 pages covering the selected project phase.

1. Design Overview

• The design phase defines the structure of ComicCraft and how its components communicate.

• The design should keep the user interface, backend logic, AI workflow, and file handling organized.

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2. System Architecture

• A simple architecture can contain a frontend layer, FastAPI backend layer, AI workflow layer, and storage/output layer.

• The frontend sends a request to the backend, which processes it and returns generated results.

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3. Module Design

• Major modules include input handling, outline generation, story generation, image generation, export, and error handling.

• Each module should have a clear responsibility.

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4. API Design

• API routes can be designed for generating an outline and generating a complete comic.

• JSON request and response formats make communication between frontend and backend simple.

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5. Data Flow Design

• Topic input flows to the outline function, then to story generation, and finally to image generation and presentation.

• The output of one stage becomes the input of the next stage.

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6. User Interface Design

• The interface should contain a topic input area, generation control, status or error message, and comic result area.

• A clean layout improves demonstration and usability.

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7. File and Folder Design

• Static folders can contain panel images and exported files.

• Python modules can contain API routes and generation functions.

• Separating generated files from source code makes maintenance easier.

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8. Error Design

• HTTP exceptions and try-except blocks can be used to handle known and unexpected failures.

• The user should receive a meaningful error response.

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9. Security and Configuration Design

• API keys should not be hard-coded in source code.

• Environment variables or secure configuration should be used for external AI services.

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10. Design Conclusion

• The design establishes a modular architecture that can be implemented step by step.

• A clear design reduces confusion during development and makes testing easier.

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